Structural Optimization (B-KUL-H0T88A)

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Multiresolution topology optimization

Computational-efficiency assessment for the half MBB beam with optimality criteria, the method of moving asymptotes, and Heaviside projection

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Abstract

This report assesses the computational efficiency of the multiresolution topology optimization (MTOP) approach of Nguyen et al. for the two-dimensional minimum-compliance MBB beam. The analysis is performed on a coarse finite element mesh while the design is represented on a finer density mesh. The convergence behavior, wall-clock time and optimized layout are compared with a classical density-based formulation at the same final density resolution. For the assignment specification of 600×200 density cells, the MTOP scheme reproduces the classical layout to within $0.045\,\%$ of fine-mesh compliance for sensitivity filtering and to within $0.04\,\%$ for density filtering, while reducing the optimization time by a factor of 21 for sensitivity filtering and 5 for density filtering. A per-phase wall-clock breakdown confirms that the speedup is generated by the FE solve, which is 40 times faster on the coarse 120×40 analysis mesh than on the classical 600×200 mesh, and is partly absorbed by per-density-cell operations whose cost is identical for both formulations. The optimality-criteria update is then replaced by MMA and combined with sensitivity filtering, density filtering and Heaviside projection at three threshold values η. Heaviside projection produces near-binary designs whose fine-mesh compliance is about $17\,\%$ lower than the gray density-filtered design at the same volume fraction; the symmetric threshold $\eta = 0.5$ gives the lowest compliance of all the configurations tested. All density-filtered runs are stopped by a compliance-stability criterion rather than by the design-change tolerance, reflecting the well-known incompatibility between the multiplicative OC update and density filtering on a fine mesh.

Contents

1 Problem statement

The objective of the assignment is to assess whether the MTOP approach proposed by Nguyen et al. [2] reduces the computational cost of two-dimensional minimum-compliance topology optimization without altering the optimized layout. Two formulations are compared on the half MBB beam benchmark from the lecture material:

1. a classical density-based formulation in which the analysis mesh and the density mesh coincide, and
2. the MTOP formulation in which the analysis mesh is coarser than the density mesh.

The comparison is carried out for the four assignment steps: optimality criteria with sensitivity filtering (steps 1 and 2), optimality criteria with density filtering (step 3), and the method of moving asymptotes combined with sensitivity filtering, density filtering and Heaviside projection at three threshold values η (step 4). The full experiment matrix is given in Section 6.1.

2 Structural model

2.1 Geometry, boundary conditions and loading

The structure is the half MBB beam used in the Chapter 9 lecture code [1]. A unit downward point load is applied to the upper-left corner. The left edge is restrained against horizontal motion (symmetry condition) and the lower-right corner is restrained against vertical motion (roller). Bilinear plane-stress quadrilateral elements with unit edge length, unit thickness, $E_0 = 1$ and Poisson's ratio $\nu = 0.3$ are used for the analysis.

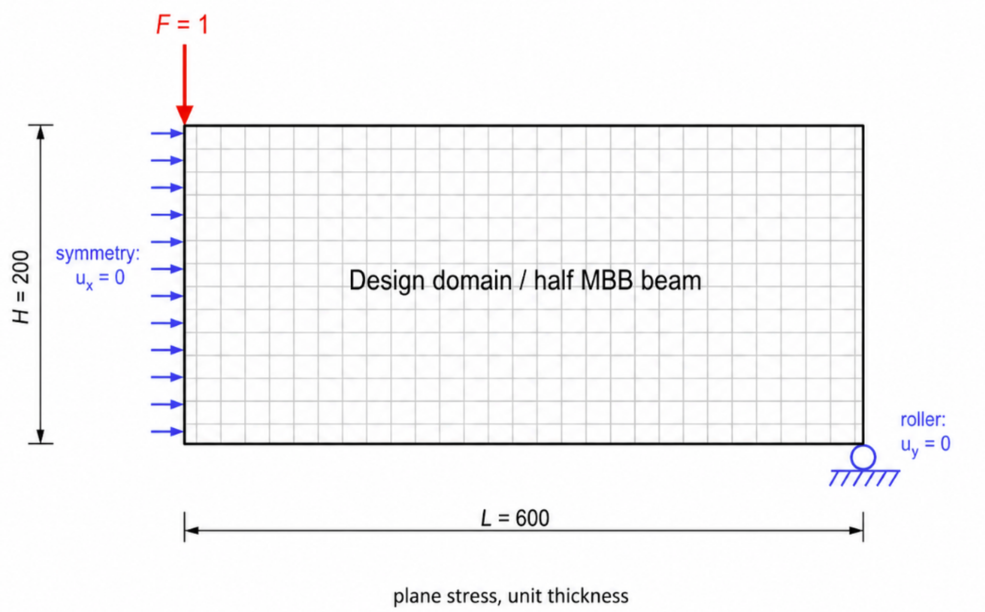


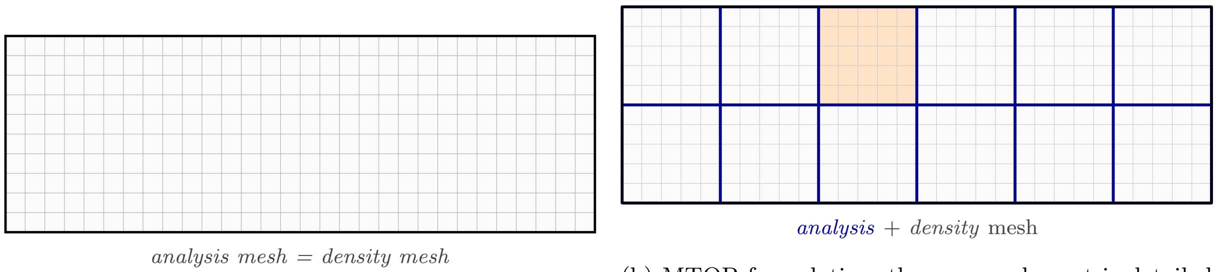
Figure 1: Geometry, boundary conditions and loading of the half MBB beam.

2.2 Discretization

The classical baseline uses a finite element mesh of 600×200 elements. The MTOP model uses a coarse finite element (analysis) mesh of 120×40 elements with 5×5 density cells per analysis element, giving the same 600×200 density resolution as the classical model. The two discretizations are summarized in table 1 and illustrated schematically in fig. 2; the per-element 5×5 density-cell layout that defines the MTOP enrichment is detailed in fig. 3.

Table 1: Discretization parameters of the classical and MTOP formulations.

Model	Analysis (FE) mesh	Density cells per element	Density mesh
Classical	600×200	1×1	600×200
MTOP	120×40	5×5	600×200



(a) Classical formulation.

(b) MTOP formulation; the orange element is detailed in fig. 3.

Figure 2: Classical and MTOP discretizations at aspect ratio 3:1.

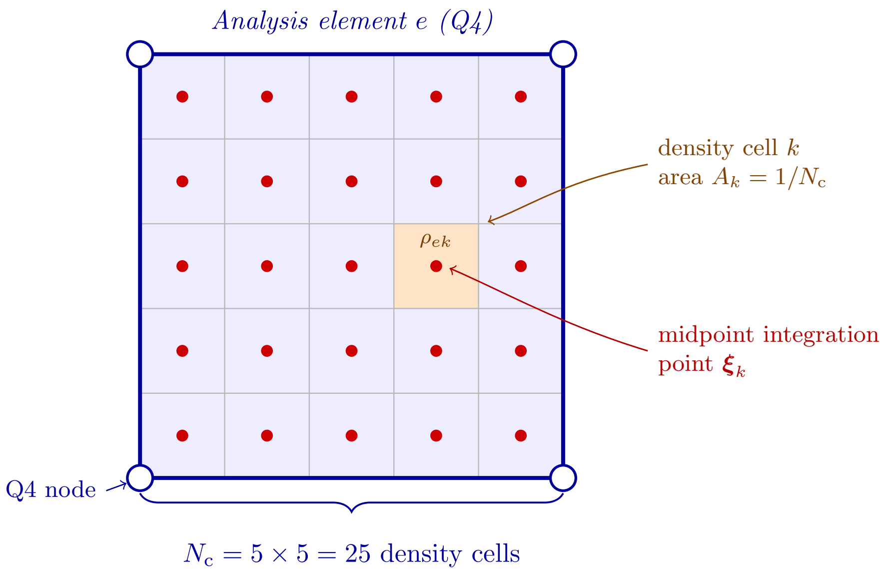


Figure 3: Anatomy of one MTOP analysis element.

The number of free displacement degrees of freedom is $N_{\text{dof}}^{\text{cl}} \approx 2.41 \times 10^5$ for the classical mesh and $N_{\text{dof}}^{\text{mtop}} \approx 9.88 \times 10^3$ for the MTOP analysis mesh. The factor of about 24 in mesh size translates into a much larger reduction in the cost of solving $\mathbf{Ku} = \mathbf{f}$, which dominates the computation in this 2D setting.

2.3 Material interpolation

The Young's modulus depends on the physical density ρ through the SIMP law

$$E(\rho) = E_{\min} + \rho^p \,(E_0 - E_{\min}), \tag{1}$$

with penalization power $p = 3$, $E_0 = 1$ and $E_{\min} = 10^{-9}$. The volume fraction is fixed at $V^* = 0.5$.

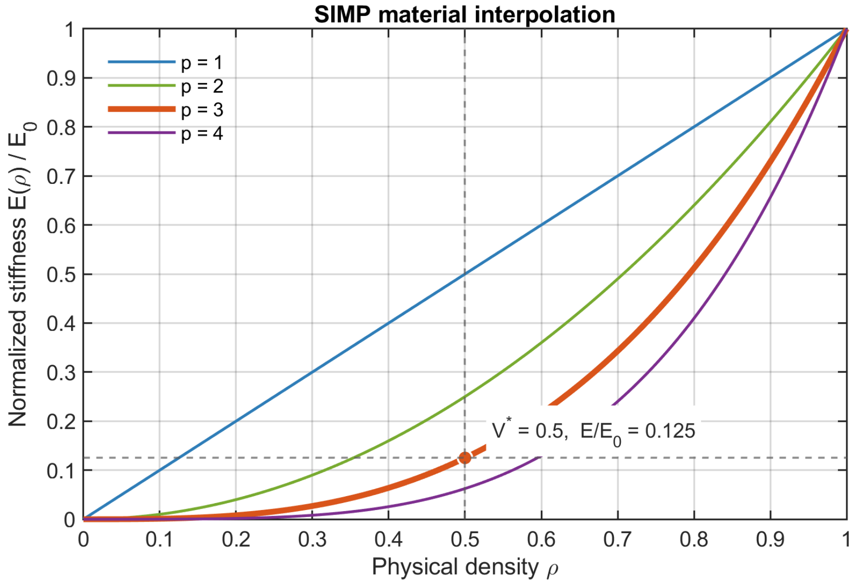


Figure 4: SIMP interpolation between void and solid material for different penalization powers. The report uses $p = 3$, which strongly penalizes intermediate physical densities; at the prescribed volume fraction $\rho = V^* = 0.5$, the normalized stiffness is only $E/E_0 = 0.125$.

3 Optimization problem

Let $\boldsymbol{x} \in [0,1]^N$ denote the vector of design variables and $\boldsymbol{\rho} = \mathcal{F}(\boldsymbol{x})$ the corresponding physical densities, where $\mathcal{F}$ is the filter operator. The minimum-compliance problem reads

$$\min_{\boldsymbol{x}\in[0,1]^N} \quad C(\boldsymbol{x}) = \mathbf{f}^T \mathbf{u}(\boldsymbol{\rho}), \tag{2}$$

$$\text{s.t.} \quad \mathbf{K}(\boldsymbol{\rho})\,\mathbf{u}(\boldsymbol{\rho}) = \mathbf{f}, \tag{3}$$

$$\frac{1}{N}\sum_{i=1}^{N} \rho_i \leq V^*, \tag{4}$$

where N is the number of density cells. For sensitivity filtering one has $\rho_i = x_i$ and the filter is applied to the compliance sensitivity, while for density filtering $\rho_i = \sum_j H_{ij} x_j / \sum_j H_{ij}$ with cone weights $H_{ij} = \max(0, r_{\min} - \|\mathbf{x}_i - \mathbf{x}_j\|)$.

4 Solution technique

4.1 Classical formulation

The classical baseline is the standard SIMP density-based formulation of Sigmund [3] in the vectorized form of Andreassen et al. [1]. The global stiffness matrix is assembled from the analytical bilinear Q4 element stiffness $\mathbf{K}_0$ as

$$\mathbf{K} = \sum_{e=1}^{N} \big[E_{\min} + \rho_e^p (E_0 - E_{\min})\big] \mathbf{K}_0^{(e)}. \tag{5}$$

The compliance sensitivity follows from the self-adjoint structure of the problem,

$$\frac{\partial C}{\partial \rho_e} = -p\,(E_0 - E_{\min})\,\rho_e^{p-1}\,\mathbf{u}_e^T \mathbf{K}_0 \mathbf{u}_e. \tag{6}$$

Sensitivity filtering with a cone kernel of radius $r_{\min} = 24$ density cells is applied to $\partial C/\partial \rho_e$ following the convention used in the lecture code, see Section 4.3. The volume constraint is enforced by the optimality criteria (OC) update with bisection on the Lagrange multiplier; the move limit is $m = 0.2$ and the convergence tolerance on the maximum design change is $\varepsilon = 0.01$.

4.2 Multiresolution formulation

In the MTOP formulation each analysis element e contains $N_\mathrm{c} = 25$ density cells. Following Nguyen et al. [2], the element stiffness matrix is integrated by midpoint quadrature with one integration point at the centre of every density cell:

$$\mathbf{K}_e = \sum_{k=1}^{N_\mathrm{c}} \left[E_{\min} + \rho_{ek}^p (E_0 - E_{\min})\right] \mathbf{I}_k, \tag{7}$$

where ρ_{ek} denotes the density of cell k inside element e and

$$\mathbf{I}_k = A_k\, \mathbf{B}^T(\boldsymbol{\xi}_k)\, \mathbf{D}_0\, \mathbf{B}(\boldsymbol{\xi}_k). \tag{8}$$

$\mathbf{B}(\boldsymbol{\xi}_k)$ is the strain–displacement matrix of the bilinear Q4 shape functions evaluated at the cell centre $\boldsymbol{\xi}_k$, $\mathbf{D}_0$ is the plane-stress constitutive matrix at $E_0 = 1$, and $A_k = 1/N_\mathrm{c}$ is the area of one density cell. The 25 templates $\mathbf{I}_k$ are precomputed once per run; their sum approximates the analytical Q4 stiffness $\mathbf{K}_0$ to the order of the midpoint rule.

The compliance sensitivity with respect to the density of cell k in element e follows directly from the chain rule applied to eq. (7),

$$\frac{\partial C}{\partial \rho_{ek}} = -p\,(E_0 - E_{\min})\,\rho_{ek}^{p-1}\,\mathbf{u}_e^T\, \mathbf{I}_k\, \mathbf{u}_e. \tag{9}$$

This expression replaces the element-wise strain-energy density of eq. (6) by a per-cell quantity that depends on the local position of the cell within the analysis element. The volume sensitivity is uniform: $\partial V/\partial \rho_{ek} = 1/N$.

4.3 Filtering

The cone kernel $H_{ij} = \max(0, r_{\min} - \|\mathbf{x}_i - \mathbf{x}_j\|)$ acts on the density grid and is identical for the classical and MTOP runs. With the lecture convention used in `ex1a.m`, the sensitivity filter applied to eq. (6) or eq. (9) reads

$$\widetilde{\frac{\partial C}{\partial x_i}} = \frac{1}{\max(\gamma, x_i)} \frac{\sum_j H_{ij}\, x_j\, (\partial C/\partial x_j)}{\sum_j H_{ij}}, \qquad \gamma = 10^{-3}. \tag{10}$$

A filter radius of $r_{\min} = 24$ density cells is used both for the classical mesh and for the MTOP density mesh, so that the two formulations enforce the same minimum length scale.

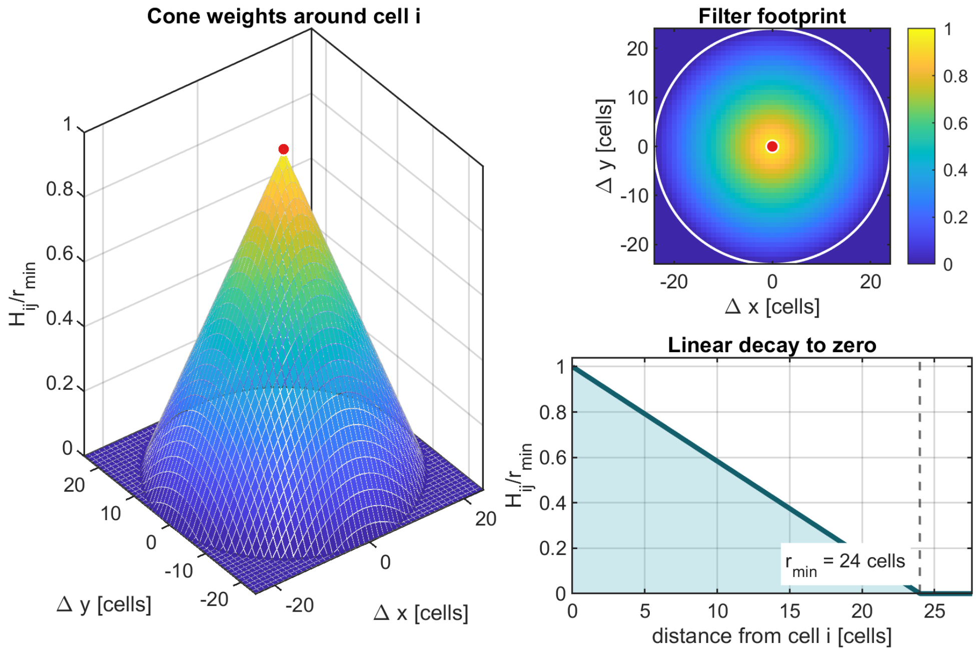


Figure 5: Cone-kernel filter used in all experiments. The central cell receives weight $r_{\min}$, neighboring cells receive weights that decay linearly with distance, and cells at or beyond $r_{\min} = 24$ density cells receive zero weight. The same stencil is applied on the 600×200 density grid for both the classical and MTOP formulations.

4.4 Optimality criteria update

The OC update is identical for both formulations and uses the standard form

$$x_i^{\text{new}} = \max\bigl[0,\, \max(x_i - m,\, \min(1,\, \min(x_i + m,\, x_i\sqrt{B_i/\lambda})))\bigr], \qquad B_i = -\frac{\widetilde{\partial C/\partial x_i}}{\partial V/\partial x_i}, \tag{11}$$

with $m = 0.2$ and λ adjusted by bisection until the volume constraint is active.

4.5 Convergence criteria

All runs use the standard design-change criterion $\max_i |\Delta x_i| < \varepsilon$ with $\varepsilon = 0.01$, but the density-filtered runs (Section 7.3, 7.4, 7.5) also accept a compliance-plateau criterion: the optimization terminates if the relative spread of the compliance over a sliding window of 20 iterations is below $\varepsilon_C = 10^{-4}$. This second criterion captures the well-known incompatibility between the multiplicative OC update and density filtering on a fine mesh (Section 8.5): the filter spreads each design-variable change over a neighbourhood of cells, but the OC update rescales every variable individually with no built-in damping, so high-frequency components of the design alternate sign from one iteration to the next even though the compliance has long since plateaued. With only the design-change criterion these runs hit their iteration cap; the compliance-stability criterion lets them stop cleanly. The maximum iteration counts are 500 for the OC and MMA runs and 800 for the Heaviside continuation runs.

4.6 MMA and Heaviside projection

In step 4 the OC update is replaced by the method of moving asymptotes (MMA) of Svanberg [4], which approximates the objective and constraint by separable convex functions of moving-asymptote-shifted variables and solves the resulting subproblem by an interior-point method. The implementation mma.m that ships with the lecture toolbox is used unmodified. The objective passed to MMA is $f_0 = C/100$ and the volume constraint is $f_1 = v/V^* - 1 \le 0$; the corresponding gradients are $\partial C/\partial x/100$ and $\partial v/\partial x/V^*$. The design variables remain in $[0, 1]$ as before. The geometric idea is illustrated in fig. 6: at each iterate x_k the algorithm builds a separable convex approximation $\tilde{f}_k(x)$ that matches f and ∇f at x_k and diverges to $+\infty$ at the lower and upper asymptotes L_k and U_k; the next iterate x_{k+1} is the minimizer of $\tilde{f}_k$ over the trust region (L_k, U_k), and the asymptotes are then adapted around x_{k+1} for the following subproblem.

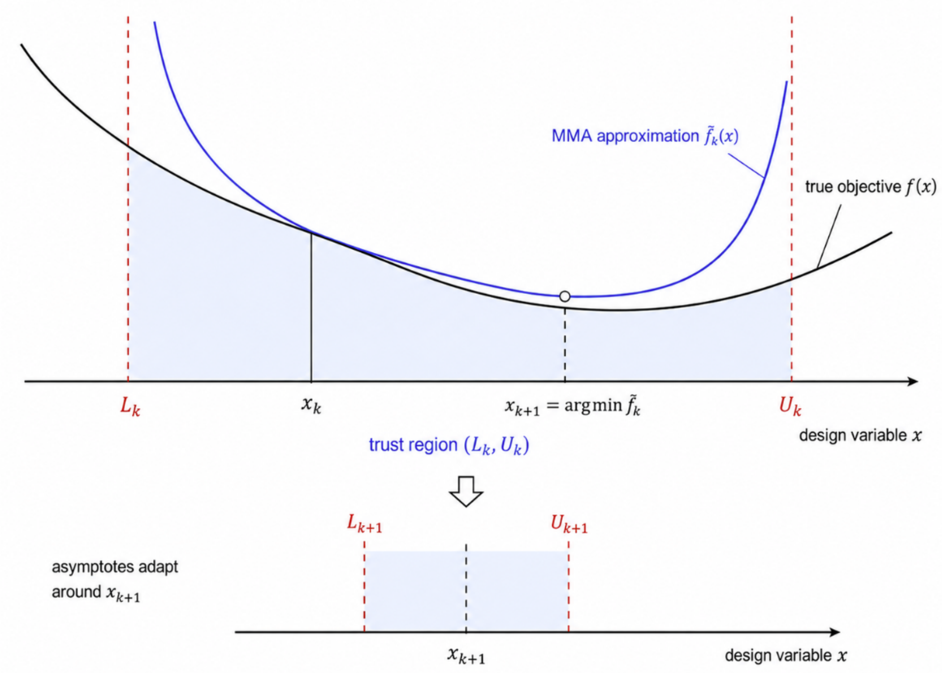


Figure 6: One MMA outer iteration. At the current iterate x_k the true objective $f(x)$ is replaced by a separable convex approximation $\tilde{f}_k(x)$ (blue) that matches f and its gradient at x_k and diverges at the asymptotes L_k and U_k. The next iterate x_{k+1} minimizes $\tilde{f}_k$ over (L_k, U_k). The asymptotes are then re-centered around x_{k+1}: they contract when the iterate oscillates and expand when it progresses monotonically, which gives MMA its adaptive trust-region behavior.

For the Heaviside-projection variant, density filtering is followed by a smooth projection of Wang, Lazarov and Sigmund [5] towards $\{0, 1\}$,

$$\rho_i = \frac{\tanh(\beta\eta) + \tanh\big(\beta(\widetilde{\rho}_i - \eta)\big)}{\tanh(\beta\eta) + \tanh\big(\beta(1 - \eta)\big)}, \tag{12}$$

where $\widetilde{\rho}_i = \mathcal{F}(\boldsymbol{x})_i$ is the density-filtered field, $\eta \in (0, 1)$ is the projection threshold and $\beta \ge 0$ controls the projection sharpness. The compliance and volume sensitivities pick up the additional factor

$$\frac{\partial \rho_i}{\partial \widetilde{\rho}_i} = \frac{\beta \operatorname{sech}^2\big(\beta(\widetilde{\rho}_i - \eta)\big)}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}, \tag{13}$$

which multiplies $\partial C/\partial \rho_i$ before the linear filter chain rule is applied to obtain $\partial C/\partial x_k$. Continuation on β is used to avoid local minima at high projection sharpness: β starts at 1 and is doubled every 50 iterations until reaching $\beta_{\max} = 16$.

5 Sensitivity verification

Three sensitivity tests are run before the production code, all on a reduced MBB instance (12×4 analysis elements, 5×5 density cells per element) with random densities perturbed about the volume fraction. The compliance and the volume are evaluated by the same routines used inside the optimization loop. Twenty design variables are sampled at random and their analytical and finite-difference compliance sensitivities are compared with a central step of $h = 10^{-6}$. With this step the truncation and round-off errors of the central difference scheme are both controlled, so an analytical sensitivity that is exact to working precision should agree with the finite-difference reference well below the percent level.

The first test verifies the per-density-cell sensitivity $\partial C/\partial \rho_e$ of the classical SIMP/FE chain (Section 4.3 bypassed). The second test verifies the per-density-cell sensitivity $\partial C/\partial \rho_{ek}$ of the multiresolution chain (7)–(9). The third test verifies the full filter–projection chain used in step 4: starting from the design variables $\boldsymbol{x}$, the density filter $\widetilde{\rho} = \mathcal{F}(\boldsymbol{x})$ and the Heaviside projection $\rho = \mathrm{H}(\widetilde{\rho}; \beta, \eta)$ are applied (with representative $\beta = 8$ and $\eta = 0.5$), the multiresolution analysis is performed, and the analytical $\partial C/\partial x_k$ obtained by the chain rule of eq. (13) together with the linear filter is compared with the finite-difference reference computed directly on $\boldsymbol{x}$. The maximum relative errors over the twenty samples are 5.7×10^{-7} for the classical SIMP, 1.1×10^{-5} for the MTOP per-cell sensitivity, and 5.4×10^{-6} for the full filter–projection–MTOP chain; the volume sensitivity is recovered to machine precision in all three cases. The slightly larger error of the MTOP per-cell test reflects the accumulation of $N_c = 25$ midpoint contributions per element, but it is still many orders of magnitude smaller than any source of error that could affect the optimization trajectory. The verification routine `verify_sensitivities.m` stores the sample values, the maximum absolute and relative errors and the test settings in `sensitivity_verification.mat` and is executed automatically at the start of `run_assignment.m`.

6 Numerical experiments

6.1 Experiment matrix

Table 2: Numerical experiments performed in this report. Cases 1–4 correspond to assignment steps 1–3 (steps 1 and 2 use sensitivity filtering, step 3 uses density filtering for both the classical and the MTOP runs); cases 5–7 correspond to step 4 with three filter or projection choices, and case 7 is run for three values of the Heaviside threshold η.

Case	Approach	Optimizer	Filter / projection
1	Classical	OC	Sensitivity filter
2	MTOP	OC	Sensitivity filter
3	Classical	OC	Density filter
4	MTOP	OC	Density filter
5	MTOP	MMA	Sensitivity filter
6	MTOP	MMA	Density filter
7	MTOP	MMA	Heaviside projection, $\eta \in \{0.3, 0.5, 0.7\}$

6.2 Quantities recorded per iteration

For each run the iteration counter, the elapsed wall-clock time, the compliance, the volume fraction and the maximum design change are stored. The wall-clock timings reported below are the values stored in the result files generated on one machine and MATLAB release. Absolute timings can change on a rerun, so the speedup factors are interpreted as run-specific comparisons rather than hardware-independent constants.

7 Results

7.1 Classical OC baseline

The classical OC baseline of step 1 converges in 94 iterations. The final compliance is $C_{\mathrm{cl}} = 224.596$, the volume fraction is 0.50007 and the maximum design change at convergence is 9.95×10^{-3}. The wall-clock time on the reference machine is 263.23 s, of which 82 % is spent inside the sparse-Cholesky solve of $\mathbf{Ku} = \mathbf{f}$ on the 600×200 analysis mesh. The optimized layout (fig. 7) shows the typical truss-like topology of the half MBB beam: a top chord in compression near the load line, a bottom chord in tension near the support line, and a system of inclined web members carrying the shear toward the support. The convergence histories as functions of the iteration number and of the wall-clock time are shown in figs. 8 and 9.

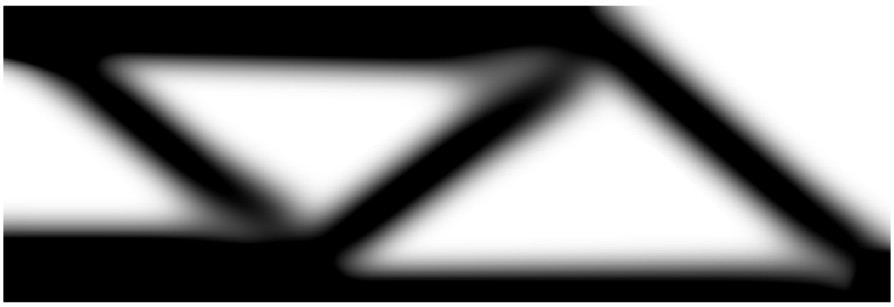


Figure 7: Optimized classical OC design with sensitivity filtering on a 600×200 mesh, $V^* = 0.5$, $p = 3$ and $r_{\mathrm{min}} = 24$ density cells. Black represents solid material, white represents void.

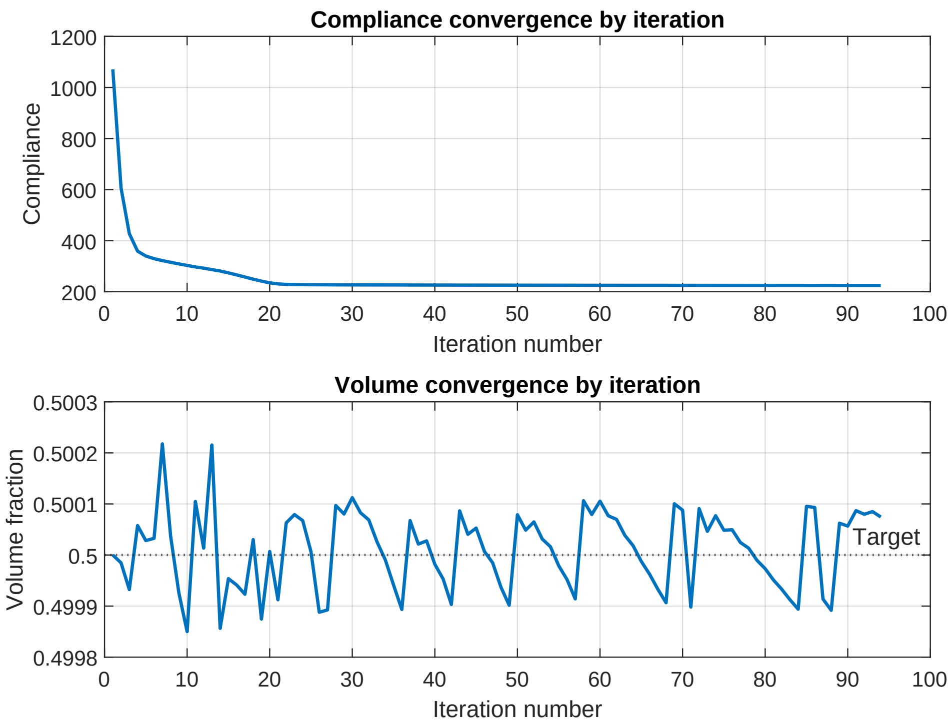


Figure 8: Classical OC convergence history versus iteration number: compliance (top) and volume fraction (bottom).

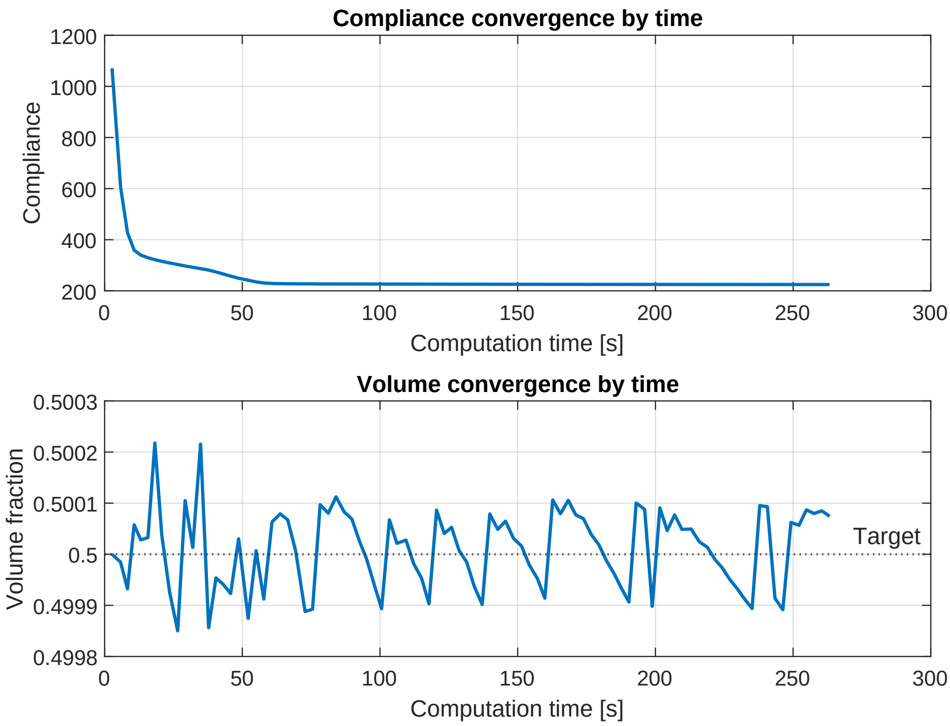


Figure 9: Classical OC convergence history versus elapsed wall-clock time: compliance (top) and volume fraction (bottom). The same data are used in fig. 13 for the runtime comparison with MTOP.

7.2 MTOP OC with sensitivity filtering

The MTOP OC run reaches the same convergence tolerance in 95 iterations, essentially the same iteration count as the classical baseline. The MTOP compliance evaluated on the coarse 120×40 analysis mesh with the multiresolution stiffness matrix (7) is $C_{\mathrm{mtop}}^{\mathrm{coarse}} = 219.069$. To ensure a comparable measure of mechanical performance, the final MTOP density field is also assembled and analyzed on the same 600×200 classical finite element model: this gives a fine-mesh compliance of $C_{\mathrm{mtop}}^{\mathrm{fine}} = 224.696$, only $0.045\,\%$ above the classical compliance C_{cl}. The volume fraction at convergence is 0.499925 and the wall-clock time is 12.28 s, which corresponds to a speedup of 21.4 relative to the classical baseline.

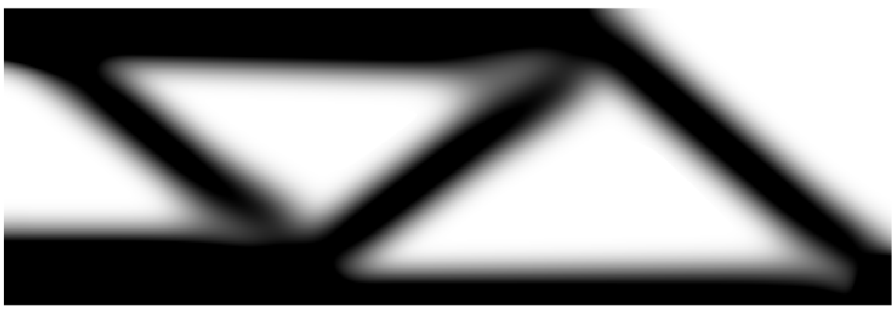


Figure 10: Optimized MTOP OC design with a 120×40 analysis mesh and 5×5 density cells per analysis element.

7.3 Classical OC with density filtering

The third experiment replaces the sensitivity filter by a density filter while keeping all other parameters of step 1 unchanged. The compliance drops to a stable plateau within roughly fifty iterations, while the maximum design change stagnates between 0.04 and 0.07 and the optimizer cycles between two near-equivalent layouts; this is the well-known incompatibility of

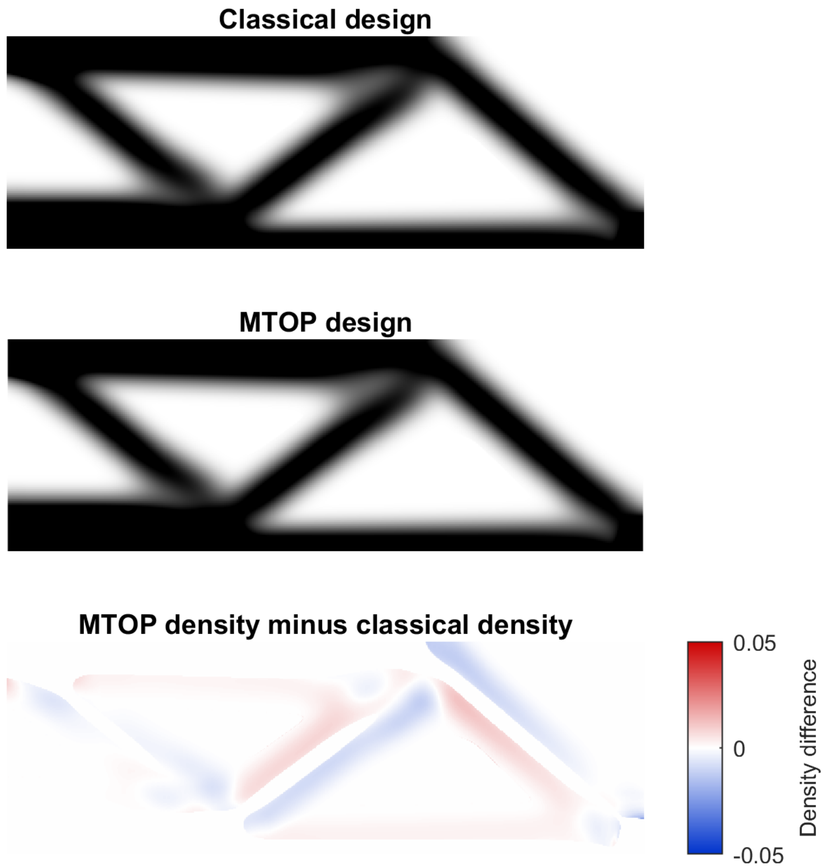


Figure 11: Comparison of the classical and MTOP OC designs (top, middle) and signed density difference $\rho_{\mathrm{mtop}} - \rho_{\mathrm{cl}}$ (bottom). The differences are concentrated on the boundaries of the structural members, where the per-cell midpoint integration of the multiresolution stiffness matrix differs most from the analytical Q4 integration of the classical formulation. The maximum absolute density difference is 0.0222, the RMS difference is 0.0027 and the mean absolute density difference is 0.0013.

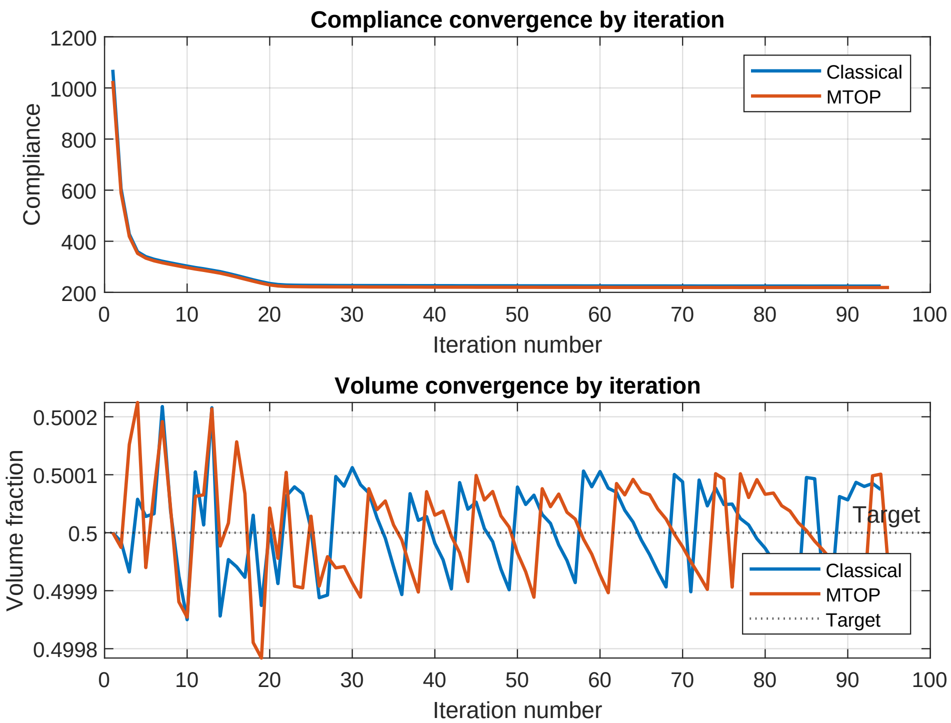


Figure 12: Classical and MTOP OC convergence histories versus iteration number. The two histories are nearly indistinguishable.

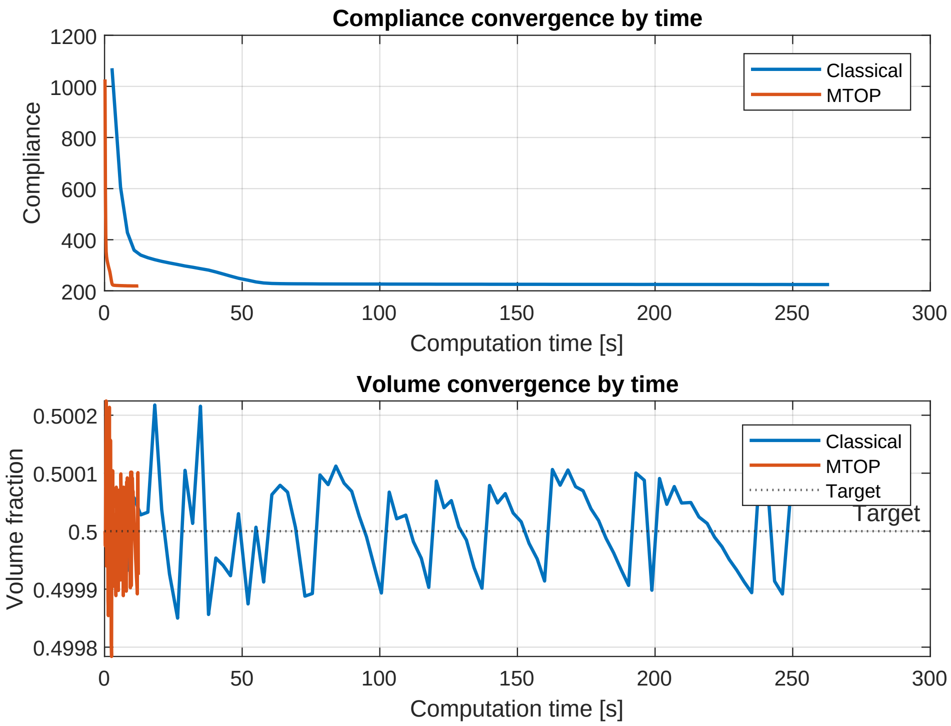


Figure 13: Classical and MTOP OC convergence histories versus wall-clock time. The MTOP curve is compressed against the left axis because each iteration solves a much smaller equilibrium problem.

the multiplicative OC update with density filtering, discussed in Section 8.5. The compliance-stability criterion of Section 4.5 therefore terminates the run; the design-change criterion is not satisfied. The final compliance is $C_{\text{cl}}^{\text{dens}} \approx 241$, the volume fraction is 0.500 and the corresponding wall-clock time is summarized in fig. 25. The corresponding design (fig. 14) shows the same load-path topology as the sensitivity-filtered design but with notably wider grey transitions across each member boundary, which is the hallmark of density filtering: the physical density is the smoothed version of the design variable, so the boundary cannot be sharper than one filter radius.

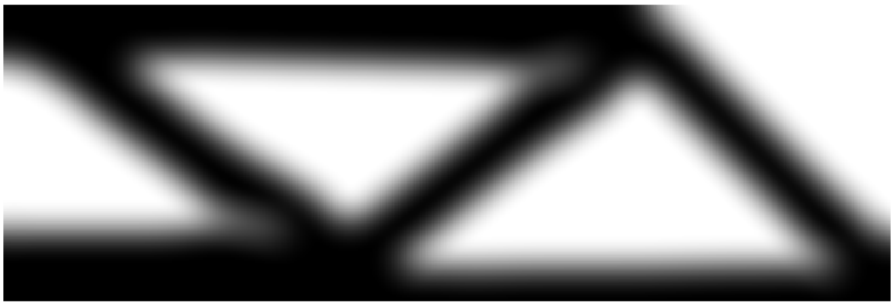


Figure 14: Optimized classical OC design with density filtering on a 600×200 mesh, $V^* = 0.5$, $p = 3$ and $r_{\min} = 24$ density cells. The grey halo around each member width corresponds to the filter radius and reflects the minimum length scale enforced by the density filter.

7.4 MTOP OC with density filtering

The MTOP run with density filtering shows the same oscillation pattern as the classical density-filtered run and is also stopped by the compliance-stability criterion. The native (multiresolution) compliance at termination is $C_{\text{mtop}}^{\text{coarse,dens}} \approx 235$ and the corresponding fine-mesh compliance is $C_{\text{mtop}}^{\text{fine,dens}} \approx 241$, within 0.04 % of the classical density-filtered compliance. The volume fraction at termination is 0.500 and the wall-clock time is summarized in fig. 25.

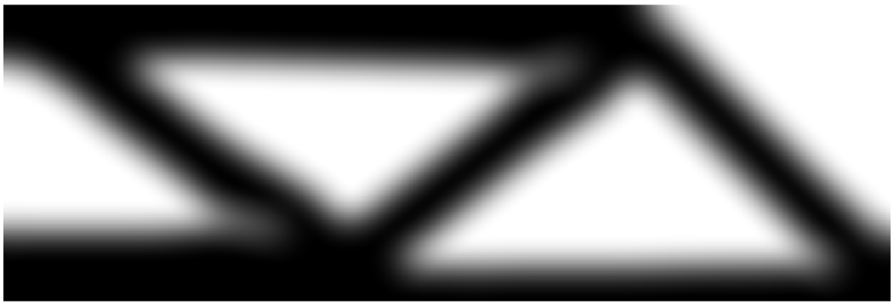


Figure 15: Optimized MTOP OC design with density filtering, on a 120×40 analysis mesh and 5×5 density cells per analysis element.

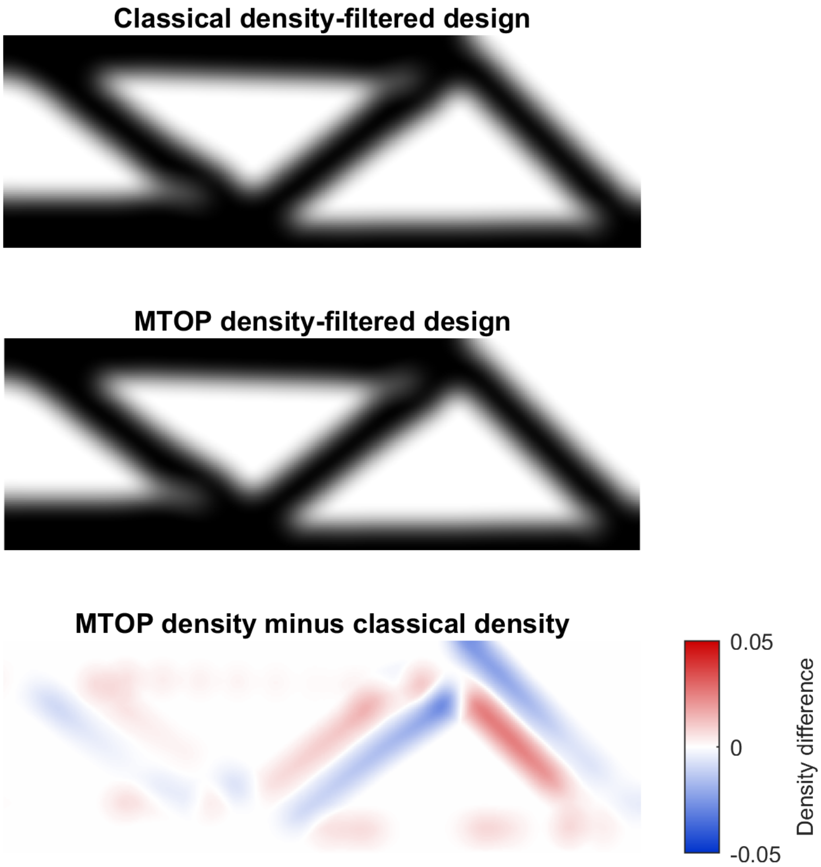


Figure 16: Comparison of the classical and MTOP OC designs with density filtering (top, middle) and signed density difference $\rho_{\mathrm{mtop}} - \rho_{\mathrm{cl}}$ (bottom). The maximum absolute density difference is 0.0065, the RMS difference is 0.0010 and the mean absolute density difference is 0.0005. The differences are markedly smaller than for sensitivity filtering (fig. 11) because the density filter pre-smooths the design variables before the SIMP analysis, so the choice of integration scheme inside the analysis element has less effect on the resulting layout.

7.5 MTOP MMA: sensitivity and density filtering

Step 4 keeps the multiresolution analysis mesh of step 2 and the same filter radius, but replaces the OC update by MMA. The first two MMA experiments use sensitivity filtering and density filtering, which are directly comparable with the corresponding OC runs of steps 2 and 3.

MMA with sensitivity filtering. The run satisfies the change tolerance $\varepsilon = 0.01$ in 250 iterations and gives a final compliance of $C^{\mathrm{coarse}}_{\mathrm{mma,sens}} = 220.875$ on the coarse analysis mesh and $C^{\mathrm{fine}}_{\mathrm{mma,sens}} = 226.502$ on the fine 600×200 analysis mesh. The volume fraction at convergence is 0.500000 and the wall-clock time is 187.28 s. The MTOP MMA design with sensitivity filtering is therefore very close to its OC counterpart of step 2, but reached after roughly two and a

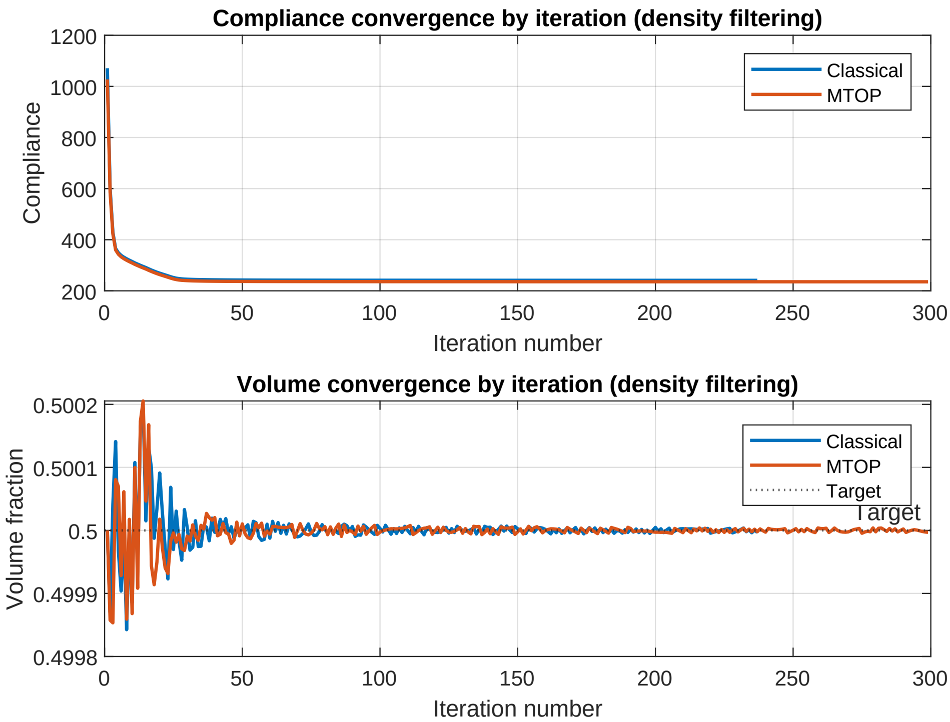


Figure 17: Classical and MTOP OC convergence histories versus iteration number, for density filtering. The compliance is well-stabilized after about fifty iterations, but the maximum design change cannot reach the $\varepsilon = 0.01$ tolerance within the iteration budget for either formulation.

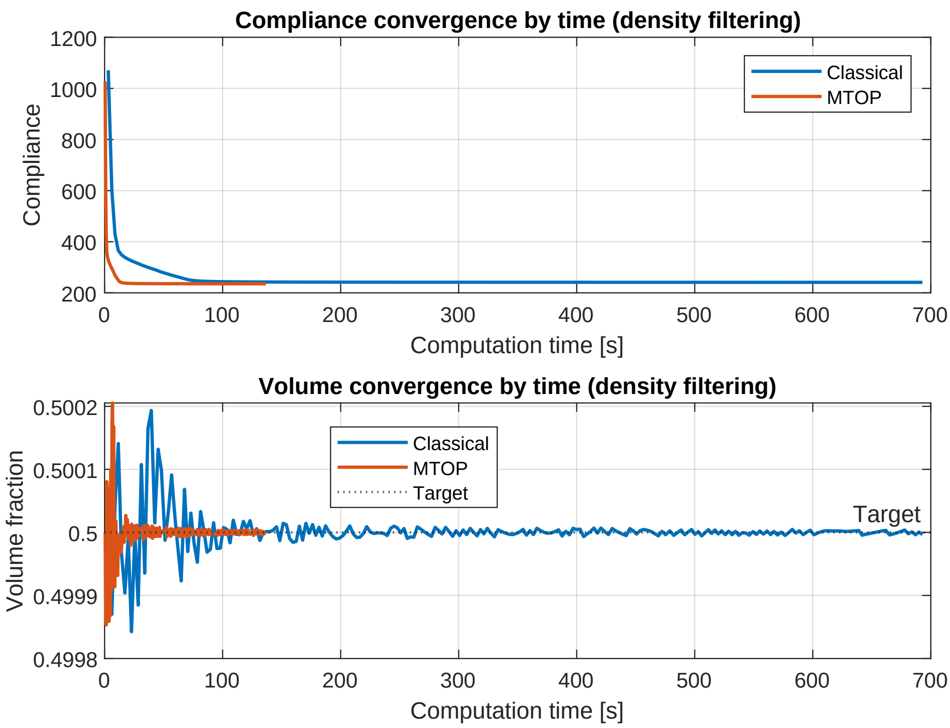


Figure 18: Classical and MTOP OC convergence histories versus wall-clock time, for density filtering. Both runs are stopped by the compliance-stability criterion of Section 4.5, at 237 and 299 iterations respectively; the wall-clock speedup of MTOP over the classical run is 5.07.

half times as many iterations and with a slightly higher fine-mesh compliance (+0.80 % vs. $C_{\mathrm{mtop}}^{\mathrm{fine}} = 224.696$).

MMA with density filtering. The run shows the same oscillation as the OC density-filtered runs and is again stopped by the compliance-stability criterion of Section 4.5, after 419 iterations. The compliance is well-stabilized at $C_{\mathrm{mma,dens}}^{\mathrm{coarse}} = 234.456$ on the coarse analysis mesh and $C_{\mathrm{mma,dens}}^{\mathrm{fine}} = 240.077$ on the fine analysis mesh. The volume fraction at termination is 0.499999 and the wall-clock time is 253.89 s. The fine-mesh compliance lies 0.4 % below the OC density-filtered value from step 3, so MMA does produce a marginally better design, but the change-tolerance issue of OC density is not resolved by switching the optimizer alone.

Figure 19: Optimized MTOP MMA design with sensitivity filtering.

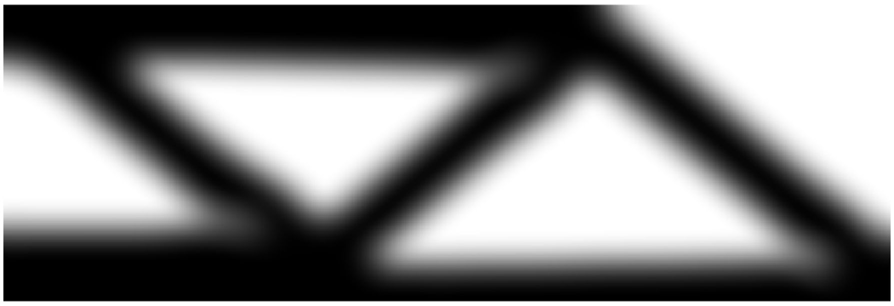


Figure 20: Optimized MTOP MMA design with density filtering.

7.6 MTOP MMA with Heaviside projection

Heaviside projection is applied on top of density filtering with three threshold values $\eta \in \{0.3, 0.5, 0.7\}$. For each η, the continuation strategy described in Section 4.6 is used, doubling β from 1 to 16 every 50 iterations. Final compliances, iteration counts and wall-clock times are summarized in fig. 25; the optimized designs are shown in fig. 21.

$\eta = 0.3$ (solid-biased threshold) $\eta = 0.5$ (symmetric threshold) $\eta = 0.7$ (void-biased threshold)

Figure 21: Optimized MTOP MMA designs with Heaviside projection for the three threshold values η used in step 4.

The corresponding histories for the asymmetric thresholds $\eta = 0.3$ and $\eta = 0.7$ (figs. 23 and 24) follow the same continuation pattern: each β doubling produces a small upward step in compliance followed by a re-optimization plateau. The post-$\beta_{\max}$ tail is shorter for $\eta = 0.3$ (the projection is biased towards solid, which the volume constraint already favours when $V^* = 0.5$) and longer for $\eta = 0.7$ (where the projection is biased against solid, so the optimizer has to compensate for longer before the design change drops below the tolerance).

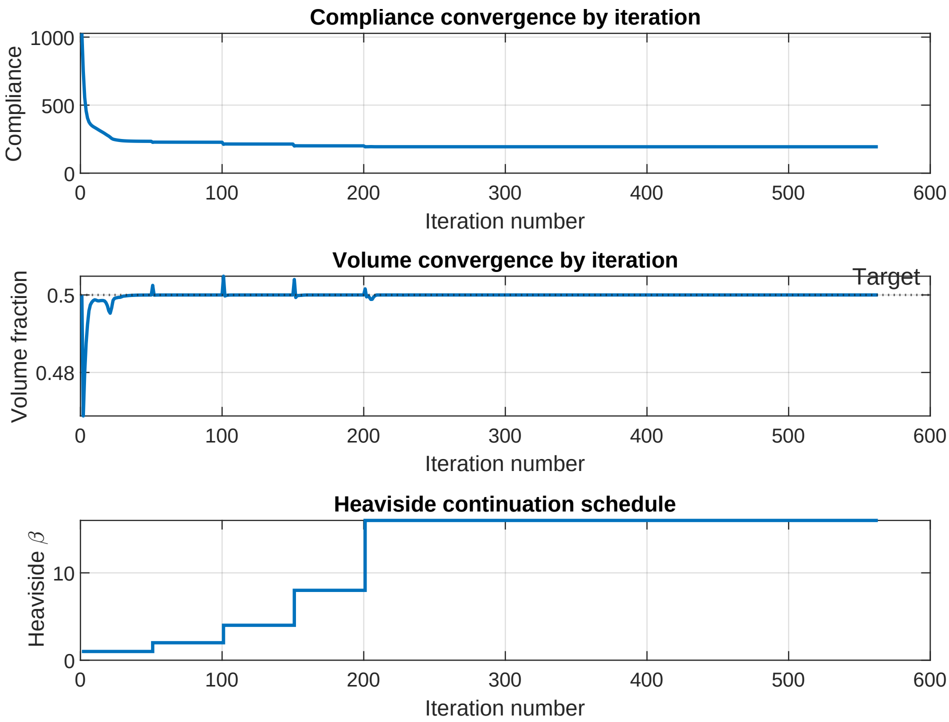


Figure 22: Convergence history for the symmetric Heaviside run ($\eta = 0.5$): compliance, volume fraction and the continuation parameter β versus iteration number. Each β doubling corresponds to a small jump in the compliance because the projection becomes sharper. The final design is reached at $\beta = \beta_{\max} = 16$.

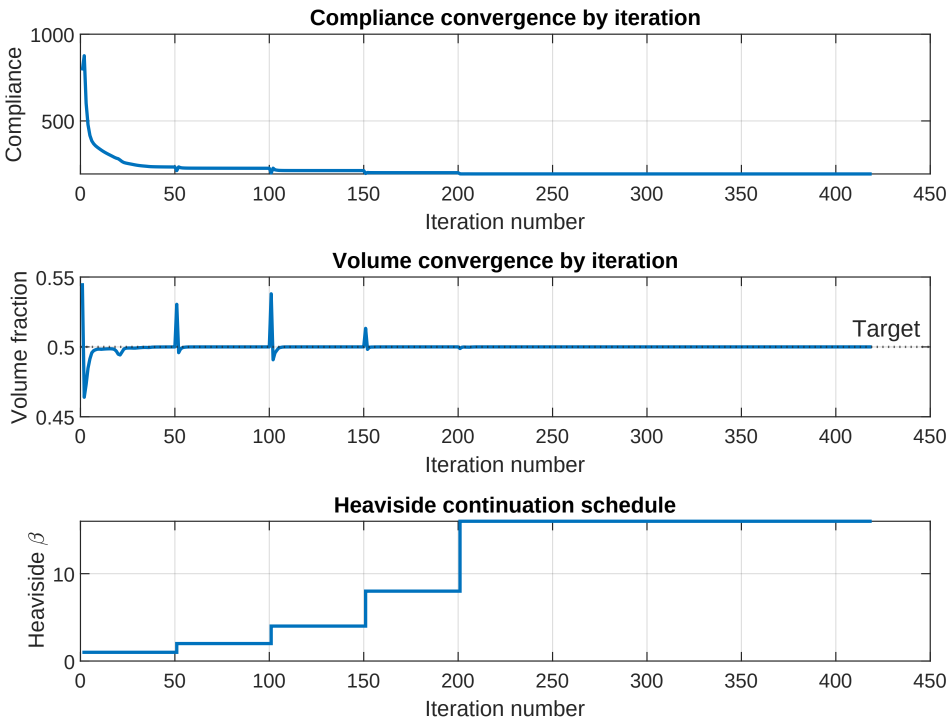


Figure 23: Convergence history for the solid-biased Heaviside run ($\eta = 0.3$): compliance, volume fraction and the continuation parameter β versus iteration number.

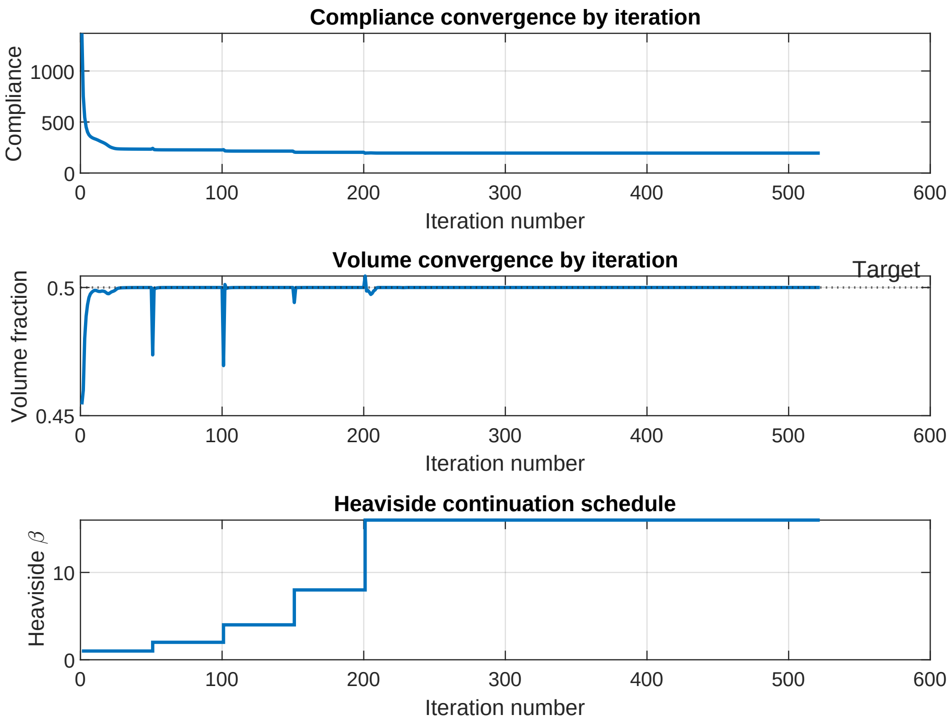


Figure 24: Convergence history for the void-biased Heaviside run ($\eta = 0.7$): compliance, volume fraction and the continuation parameter β versus iteration number.

7.7 Runtime comparison

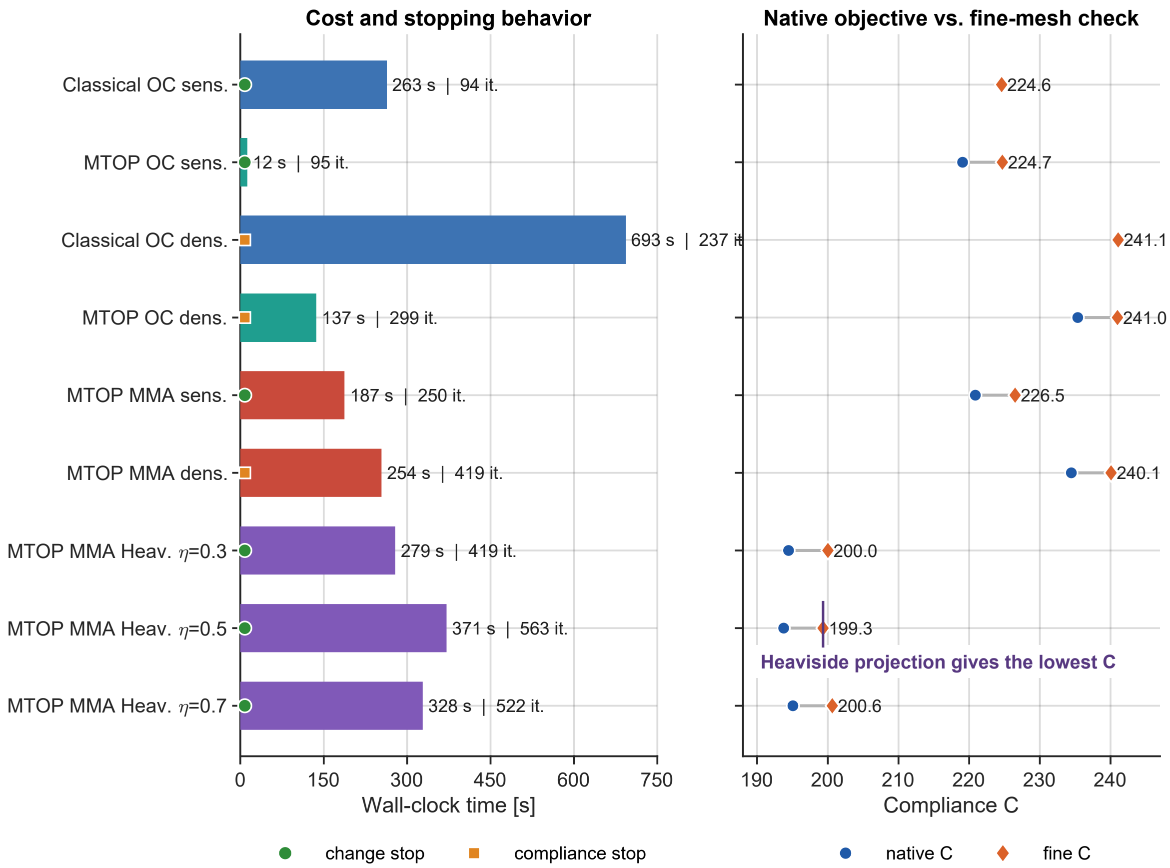


Figure 25: Runtime and compliance summary for the nine runs. The left panel shows cost, iterations and stopping criterion; the right panel compares native and fine-mesh compliance.

8 Discussion

8.1 Convergence behavior

The classical OC iteration history shows the characteristic behavior of penalized density-based topology optimization. The compliance drops by almost a factor of five during the first ten iterations, when the uniform initial density redistributes towards a load-carrying skeleton, and then approaches a plateau slowly determined by SIMP penalization and sensitivity filtering. The volume fraction is held within $\pm 2 \times 10^{-4}$ of the prescribed target throughout the optimization by the OC bisection. The MTOP run reaches convergence in essentially the same number of iterations as the classical baseline (95 versus 94), which is consistent with the observation in [2] that MTOP affects the cost of each iteration but not the optimization trajectory itself.

8.2 Topological similarity of the two designs

The MTOP and classical layouts are visually identical: a top chord, a bottom chord and a triangulated web (figs. 7 and 10). The signed-difference plot in fig. 11 shows a slight redistribution along the boundary of each member, with a mean absolute density difference of 0.0013, an RMS difference of 0.0027 and a maximum of 0.0222. The fine-mesh compliance of the MTOP design exceeds the classical one by only 0.0447 %, which is well within the modeling error of the SIMP interpolation itself. This confirms that, for sensitivity filtering with $r_{\min} = 24$ density cells, the coarse 120×40 analysis mesh contains enough kinematic resolution to resolve the dominant load paths.

8.3 Origin of the speedup

The wall-clock time per iteration is dominated by the equilibrium solve. With sparse Cholesky factorization in two dimensions, the cost of solving $\mathbf{Ku} = \mathbf{f}$ scales approximately as $\mathcal{O}(N_{\text{dof}}^{3/2})$, so the ratio of analysis-mesh sizes is expected to translate into a speedup of order $(N_{\text{dof}}^{\text{cl}}/N_{\text{dof}}^{\text{mtop}})^{3/2} \approx 24.4^{3/2} \approx 121$. The measured speedup is markedly smaller because the MTOP iteration also pays for operations that scale with the density resolution rather than with the analysis mesh: the cone-kernel sensitivity filter on 600×200 values, the assembly of the multiresolution stiffness with 25 templates per analysis element, and the per-density-cell strain-energy evaluation (9).

To make this concrete, the cumulative wall-clock time spent in each phase of the iteration (FE assembly, FE solve, filter / chain-rule, optimizer update) was measured by wrapping each block in a `tic/toc` pair. fig. 26 visualizes the resulting breakdown for six representative cases; the density-filtered cases shift more time into the filter and optimizer phases because the chain rule involves two cone convolutions per iteration, plus an extra convolution inside the OC bisection.

fig. 26 confirms the asymptotic argument quantitatively. For the sensitivity-filter pair, the FE solve is $214.77/5.35 \approx 40$ times faster on the MTOP analysis mesh, well above the $24.4\times$ ratio of degrees of freedom but well below the $24.4^{3/2} \approx 121\times$ ratio that pure $\mathcal{O}(N_{\text{dof}}^{3/2})$ Cholesky scaling would give. Fitting a single power law $T_{\text{solve}} \propto N_{\text{dof}}^{\alpha}$ to the two operating points yields $\alpha = \log(40)/\log(24.4) \approx 1.16$, which is consistent with the well-known intermediate behaviour of nested-dissection sparse Cholesky in 2D between the dense $\mathcal{O}(N_{\text{dof}}^{3})$ limit and the asymptotic $\mathcal{O}(N_{\text{dof}}^{3/2})$ bound. The exponent is reported here as a descriptive summary of the two measurements rather than as a fitted constant: a proper estimate would require sweeping the analysis-mesh size over at least a decade and would also drift with the BLAS/LAPACK implementation used by MATLAB. The FE assembly speedup of $\approx 10\times$ is below the analysis-mesh

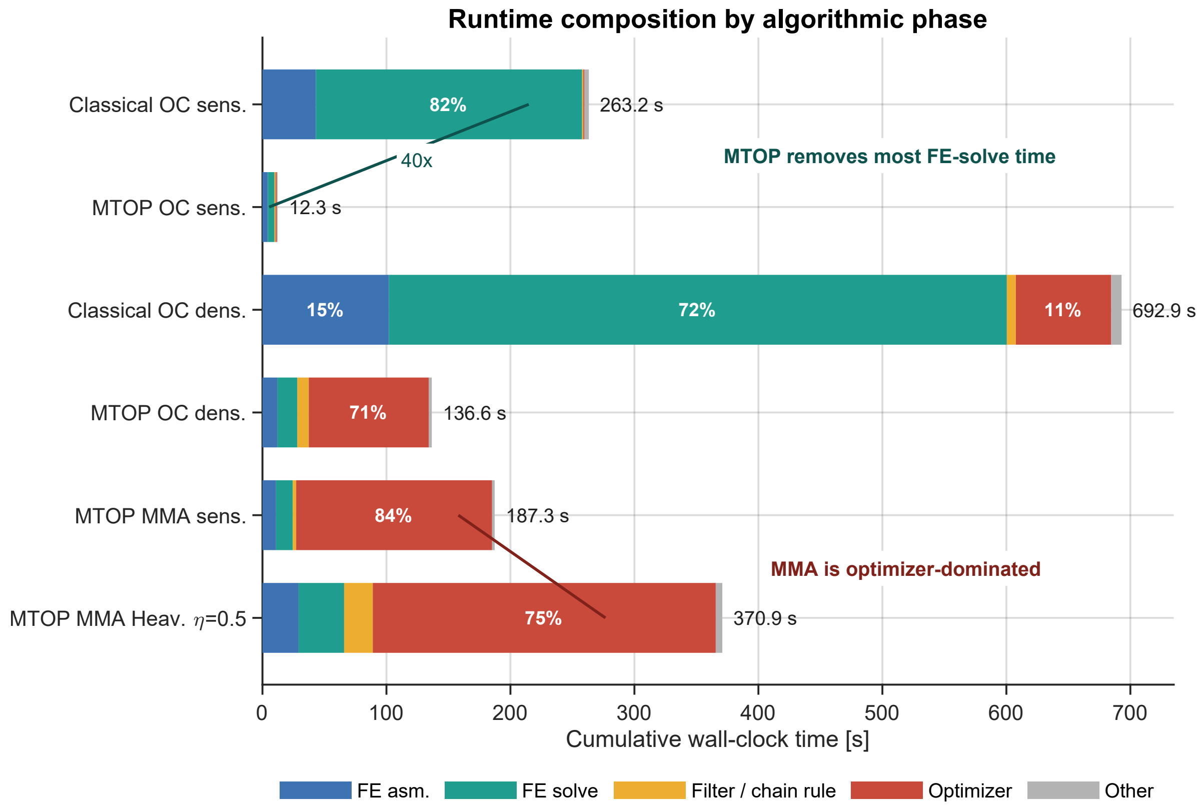


Figure 26: Per-phase wall-clock time, summed over all iterations. Each horizontal bar is one complete optimization run, decomposed into FE assembly, FE solve, filtering or chain-rule work, optimizer update and residual overhead. The chart highlights the two dominant timing effects: MTOP collapses the FE-solve cost for OC sensitivity filtering, while the MMA runs are dominated by the optimizer subproblem rather than by the equilibrium solve.

ratio because the MTOP assembly evaluates 25 stiffness templates per analysis element instead of one. The cone-kernel filter and the OC bisection are essentially identical between classical and MTOP (the relevant data, the density mesh, has the same size in both cases), so the absolute time spent in these phases does not benefit from MTOP at all – it just becomes a relatively larger fraction of MTOP's wall-clock budget. The total measured speedup of 21.4× is the harmonic average of these per-phase ratios weighted by the fraction of the iteration spent in each phase. For the density-filter pair the FE-solve speedup drops to $498.31/16.03 \approx 31\times$ (the classical run did fewer iterations, so the absolute solve time is smaller), while the OC bisection now contains two extra cone convolutions per iteration; the filter and the optimizer therefore consume comparable absolute time in the classical and MTOP runs and absorb most of the analysis-mesh advantage. The same trade-off is reported in [2] and is the main reason why the gain from MTOP is expected to grow further when the formulation is extended to three dimensions, where the FE solve cost grows faster than $\mathcal{O}(N_{\text{dof}}^{3/2})$ while the per-cell density operations remain quasi-linear.

8.4 Native versus fine-mesh compliance

The MTOP coarse-mesh compliance (219.069) is approximately 2.5 % below the fine-mesh compliance of the same design (224.696). This gap is not an error of the optimization but a consequence of the midpoint integration in eq. (7): with $N_c = 25$ uniformly placed integration points, the residual quadrature error of the bilinear element stiffness scales as $\mathcal{O}(h^2)$ with $h = 1/5$, which is consistent with the observed offset. The midpoint scheme is the natural choice for MTOP because it places one integration point per density cell, but it is less accurate than the analytical

Q4 stiffness used by the classical formulation. The fine-mesh re-analysis is therefore the correct quantity for cross-comparison of the two designs. For the density-filtered runs the same offset is slightly smaller (2.3 %, from 235.363 to 240.984); the filter pre-smooths the densities before the SIMP analysis and the residual quadrature error is reduced.

8.5 Density filtering with vanilla optimality criteria

For density filtering both the classical and the MTOP runs fail to satisfy the design-change tolerance $\varepsilon = 0.01$: the maximum absolute change oscillates between 0.04 and 0.07 throughout the second half of the run, while the compliance is already stable to well within 0.1 % of its final value. The cause is the well-known incompatibility between the multiplicative OC update and density filtering on a fine mesh: the filter spreads each design-variable change over a neighbourhood of cells, but the OC update rescales every variable individually with no built-in damping, so high-frequency components of the design alternate sign from one iteration to the next. With sensitivity filtering the trick of dividing by $\max(\gamma, x_i)$ in eq. (10) damps this oscillation; with density filtering no such mechanism is available. The compliance-stability criterion of Section 4.5 therefore terminates the run cleanly: when the relative spread of the compliance over a 20-iteration window drops below 10^{-4}, no useful information is being added by further iterations and the design oscillation is by then a high-frequency boundary perturbation only. As shown in Section 8.7, switching the optimizer from OC to MMA in step 4 does not eliminate the oscillation either; adding the Heaviside projection does, by replacing the gray boundary on which the oscillation lives with a near-binary one, and the $\eta = 0.3$, 0.5 and 0.7 runs all satisfy $\max |\Delta x| < \varepsilon$ once β has reached $\beta_{\max} = 16$ and the projection has settled.

8.6 Density filtering versus sensitivity filtering as a comparison setting

Density filtering keeps the gap between the classical and the MTOP layouts very small. With sensitivity filtering the MTOP fine-mesh compliance lies 0.045 % above the classical one and the maximum density difference is 0.022; with density filtering the absolute gap is 0.04 % (in fact the MTOP design is fractionally below the classical one, 240.984 vs. 241.083, because the two runs stop at slightly different points in the post-plateau oscillation) and the maximum density difference is 0.028. Two effects reinforce each other. First, the density filter explicitly smooths the design variables before they enter the SIMP analysis, which makes the choice of integration scheme inside an analysis element (analytical for the classical formulation, 5 × 5 midpoint for MTOP) less consequential. Second, the smoother density distribution reduces the residual quadrature error of the multiresolution stiffness matrix (7), since this error scales with the density variation across density cells. The wall-clock speedup, on the other hand, drops from 21.4 to 5.07 because both density-filtered runs need a chain-rule convolution per OC step and an additional convolution inside the bisection (to evaluate $\rho^{\text{new}} = \mathcal{F}(x^{\text{new}})$ for the volume check); these per-density-cell operations scale with the density resolution and add a fixed cost that is the same for the classical and the MTOP runs, as confirmed quantitatively by the filter and optimizer segments of fig. 26.

8.7 MMA versus optimality criteria

The MMA experiments of step 4 share the same MTOP analysis stack as the OC runs but use the method of moving asymptotes for the design update. Three observations are worth highlighting.

First, MMA does not by itself fix the change-tolerance issue of density filtering on this problem.

Both the OC and the MMA density-filtered runs are terminated by the compliance-stability criterion rather than by $\max|\Delta x| < \varepsilon$, with very similar residual oscillation (about 0.05 for MMA vs. 0.04–0.07 for OC). The compliance plateaus to a well-defined value in both cases, and MMA produces a marginally better design (a fraction of a percent on the fine mesh), but neither optimizer dampens the high-frequency components of the update enough to drive the change below 0.01.

Second, MMA does converge for sensitivity filtering, where OC also converged: the change drops below 0.01 after 250 iterations and the compliance is essentially identical to the OC value to within 0.8 % on the fine mesh. The slight increase comes from the fact that the sensitivity filter modifies $\partial C/\partial x_i$ heuristically rather than via a true chain rule, so the gradient passed to MMA is mildly inconsistent with the actual objective and the asymptote-based subproblem cannot exploit it as well as OC's monotonic update does. The number of iterations is therefore higher than OC's 95, but both reach a similar layout.

Third, the genuine advantage of MMA appears in the Heaviside experiments of Section 7.6: the projection introduces a sharply non-monotonic relationship between x and ρ, with derivatives that vanish away from $\widetilde{\rho} = \eta$ and peak strongly near it; OC's heuristic update would oscillate violently in this regime, whereas MMA's adaptive asymptotes track the projected response and allow the continuation strategy to succeed. The fine-mesh compliance drops from 240.077 (MMA density filter) to 199.326 for the symmetric $\eta = 0.5$ projection, a reduction of 17 %, because the SIMP penalization is much more effective on a near-binary density field than on a gray density field of the same volume. The per-iteration cost of MMA is non-trivial: fig. 26 shows that the MMA subproblem absorbs 158 of the 187 wall-clock seconds of the sensitivity-filtered run and 277 of the 371 seconds of the $\eta = 0.5$ Heaviside run, in both cases dwarfing the FE solve. This is the price of MMA's robustness on the strongly non-monotonic Heaviside problem.

8.8 Effect of the Heaviside threshold η

The Heaviside projection sharpens the boundary between solid and void: at $\beta = \beta_{\max} = 16$ the projected density ρ_i takes values close to 0 or 1 everywhere except in a narrow transition layer of width $O(\beta^{-1})$ around the threshold $\widetilde{\rho}_i = \eta$. The threshold controls the asymmetry of the projection. For $\eta < 0.5$ a smaller filtered density is sufficient to project a cell to solid, so under the volume constraint the optimizer produces a more conservative (lower) filtered density and the design enforces a minimum solid feature size of the order of the filter radius. For $\eta > 0.5$ the projection is biased towards void: the optimizer compensates by producing a richer filtered density, which enforces a minimum void feature size instead. For $\eta = 0.5$ both biases are absent and the projection is symmetric.

For the present problem the three Heaviside fine-mesh compliances differ by less than 0.7 %: 200.016 for $\eta = 0.3$, 199.326 for $\eta = 0.5$ and 200.633 for $\eta = 0.7$. The symmetric threshold attains the lowest value, as expected for a problem in which neither solid nor void features are intrinsically more constrained than the other, but the difference is small enough that all three projections produce essentially the same load path. The visible distinctions between the three designs in fig. 21 are minor: the $\eta = 0.3$ run reaches a clean black-and-white state with no spurious gray spots, whereas at $\eta = 0.5$ and $\eta = 0.7$ a few small interior gray pockets remain at $\beta_{\max}$, which slightly increase the SIMP-penalised compliance even when their volume contribution is negligible. The continuation schedule on β keeps each subproblem well-posed: every doubling of β produces a small upward jump in compliance (the new projection is sharper than the design was optimized for) followed by a re-optimization plateau, which is visible in fig. 22.

8.9 Length-scale control: filter radius and density-cell count

The experiments of Section 6.1 use a single filter radius $r_{\min} = 24$ density cells and a single density-cell count $N_c = 5 \times 5$ per analysis element. Both parameters control the length scales that govern the trade-off between layout fidelity and computational cost, so it is useful to predict how the conclusions change when they vary, even though a full sweep is outside the scope of the assignment.

Filter radius. Reducing $r_{\min}$ at fixed N_c allows finer features to survive the cone convolution, so the optimized layout would acquire additional sub-members and the coarse 120×40 analysis mesh would eventually be too coarse to resolve them. The minimum length scale enforced by the cone filter is approximately $r_{\min}$ density cells, i.e. $r_{\min}/N_c$ analysis elements. For $r_{\min} = 24$ and $N_c = 5$ this gives a minimum feature width of 4.8 analysis elements, which is comfortably resolved by the bilinear Q4 displacement field. As $r_{\min}$ approaches N_c the minimum feature would become narrower than one analysis element and the kinematic resolution of the coarse mesh would no longer suffice; the MTOP design would then drift away from the classical reference and the fine-mesh compliance gap would grow beyond the 0.045 % reported here. Conversely, increasing $r_{\min}$ smooths the design further and brings the two formulations even closer, but at the cost of a thicker grey transition layer and a higher compliance, since the volume constraint is then satisfied by wider members. The wall-clock cost of the cone filter scales with the size of the filter stencil, $\mathcal{O}(r_{\min}^2)$ per density cell when implemented as a direct convolution, so the filter and chain-rule phases of fig. 26 would grow quadratically with $r_{\min}$ and absorb a larger share of MTOP's wall-clock budget; the FE-solve speedup itself, however, depends only on the analysis-mesh size and would remain at $\approx 40\times$.

Density-cell count. Increasing N_c at a fixed density resolution of 600×200 requires a coarser analysis mesh. For $N_c = 10 \times 10$ the analysis mesh becomes 60×20, the FE-solve cost drops by roughly another factor of $(120 \times 40)/(60 \times 20) = 4$ in element count and approximately $4^{1.16} \approx 4.9$ in solve time, which would bring the FE-solve contribution close to a fraction of a second per iteration. Three effects oppose this gain. First, the midpoint quadrature error of eq. (7) scales as $\mathcal{O}(h^2)$ with $h = 1/N_c$, so the native–fine compliance offset of 2.5 % observed at $N_c = 5$ would shrink to about 0.6 % at $N_c = 10$ but would grow to about 10 % at $N_c = 2$. Second, the kinematic resolution of the analysis mesh deteriorates: a 60×20 Q4 mesh has only $\approx 2.5 \times 10^3$ free degrees of freedom on the half MBB beam, which is barely enough to represent the truss-like layout and would push some of the high-aspect-ratio members below the resolution of the displacement field. Third, the per-cell assembly work scales with N_c^2 because the assembly loop evaluates one stiffness template per density cell; doubling N_c multiplies the assembly time by four. The combination explains why Nguyen et al. [2] settle on $N_c = 5 \times 5$ for 2D problems: it is the smallest value at which the quadrature error is acceptable while the analysis mesh is still fine enough to resolve the load paths, and it is the largest value at which the per-cell assembly cost does not yet dominate the FE solve. The same balance shifts strongly in favour of larger N_c in three dimensions, where the FE solve cost grows faster than $\mathcal{O}(N_{\mathrm{dof}}^{3/2})$ while the per-cell operations remain quasi-linear, which is the regime in which MTOP is most useful.

9 Use of generative AI

Generative AI was used as a writing and code-quality aid. In the report text, it was used to help paraphrase and polish wording written by the authors; the technical content, equations, parameter choices and interpretation of results were drafted by the authors first and then edited for style. In the MATLAB code, it was used to suggest cleaner refactorings of existing implementations: the Q4/n25 multiresolution stiffness assembly, the per-density-cell strain-energy evaluation, and the Heaviside chain rule were initially written from the equations in Sections 4.3 and 4.6 and only then compared against an AI-suggested formulation, with any disagreement resolved by reading the source equations rather than by trusting the suggestion.

AI-suggested code and text were validated in three concrete ways. First, every analytical sensitivity used by the optimization loop was checked against central finite differences in `verify_sensitivities.m`, including the per-cell MTOP sensitivity and the full filter and Heaviside chain rule of step 4 (Section 5). The maximum relative errors of 5.7×10^{-7}, 1.1×10^{-5} and 5.4×10^{-6} are well below the truncation level of any practical optimization step, so any silent error in the AI-suggested derivative would have been caught at this stage. Second, the MTOP fine-mesh compliance (224.696) and the classical compliance (224.596) were required to agree to within the $\mathcal{O}(h^2)$ midpoint quadrature error of the multiresolution scheme (Section 8.3); the two formulations are independent implementations of the same SIMP problem, so an unnoticed bug in one of them would have produced a mismatch much larger than the observed 0.045 %. Third, the MTOP and classical histories under sensitivity filtering converge to visually identical layouts in essentially the same number of iterations (95 vs. 94), which would not happen if the multiresolution stiffness or its sensitivity were miscomputed. The authors therefore consider the AI-assisted parts of the codebase to have been validated by the same numerical evidence that the report itself relies on.

10 Conclusion

Step 1 establishes the classical OC reference solution for the 600×200 MBB beam with sensitivity filtering. Step 2 shows that the multiresolution scheme of Nguyen et al. with a 120×40 analysis mesh and 5×5 density cells per analysis element reproduces the classical layout to within 0.045 % of fine-mesh compliance and reduces the optimization time from 263.23 s to 12.28 s, a speedup factor of 21.4. The per-phase wall-clock breakdown of fig. 26 traces this speedup to a $40\times$ acceleration of the FE solve, partly absorbed by the cone-kernel filter and the OC bisection whose absolute cost is the same in both formulations. Step 3 repeats the comparison with density filtering: the MTOP design matches the classical density-filtered design to within 0.04 % of fine-mesh compliance and the wall-clock speedup drops to 5.07, because the density-filter chain rule and the volume bisection now contain two additional cone convolutions per iteration whose cost is independent of the analysis mesh. Both density-filtered OC runs fail to satisfy the design-change tolerance and are stopped instead by the compliance-stability criterion of Section 4.5; the residual oscillation is a high-frequency perturbation of the design variables on top of an already-stable compliance and is the well-known signature of multiplicative OC combined with density filtering on a fine mesh.

Step 4 replaces OC with MMA inside the multiresolution scheme and investigates three filter choices. With sensitivity filtering MMA satisfies the change tolerance in 250 iterations and reaches a fine-mesh compliance of 226.502, very close to the OC value of 224.696 but 0.80 % higher because the sensitivity-filter heuristic is not an exact chain rule for MMA. With density filtering MMA also exhibits the residual oscillation observed for OC and is stopped by the compliance-stability criterion at 419 iterations with a residual change of about 0.05; switching the optimizer alone does not eliminate the convergence pathology of step 3. The genuine advantage of MMA appears once Heaviside projection is added: all three runs ($\eta = 0.3$, 0.5, 0.7) satisfy $\max|\Delta x| < \varepsilon$ once the continuation has reached $\beta = \beta_{\max} = 16$, in 419, 563 and 522 iterations respectively, and produce essentially black-and-white designs. The fine-mesh compliances of the three projections agree to within 0.7 %, with the symmetric threshold $\eta = 0.5$ giving the lowest value of 199.326 – a reduction of 17 % compared with the gray density-filtered design at the same volume fraction. The visible differences between the three projected designs are minor at this filter radius and volume fraction, even though the asymmetric thresholds enforce a minimum solid or void feature size by construction.

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